GEOS NT Development



1 System Requirements 3

2 Installation and Setup 3

2.1 Installation Details 4

2.2 NT Configuration 6

2.3 Environment Configuration 7

2.4 Registry Configuration 7

3 Launching NT GEOS and Swat 9

4 Compiling GEOS Applications 11

4.1 Changes From the DOS SDK 12

5 Directory Structure 13

5.1 Branches 13

5.2 Main Directories 15

5.3 Target Directories 15

5.4 Developing In Your Working Branch 16

5.5 Multi-Developer Projects 17

6 Troubleshooting 17





Welcome to GEOS development in the Windows NT environment. This document guides you through the CD-ROM installation procedure; it also describes your environment configuration, shows how to set up multiple NT machines, and lists the most common setup difficulties in a short troubleshooting guide.

In order to complete the SDK installation and compile geodes, you need a perl interpreter; your CD-ROM includes a shareware copy. In order to compile, you need both a perl interpreter and the Borland C/C++ compiler, version 3.1 or later.

If you are familiar with GEOS development, please review the SDK changes described in "Compiling GEOS Applications," Section 4, and "Directory Structure," Section 5, before you begin compiling and debugging geodes.

You can find complete information on the tools mentioned here (Swat, pmake, etc.) in the Tools book.

1 System Requirements

Hardware	Pentium 133 with 32 megs of RAM recommended.
Software	Windows NT, version 3.51 or later. In addition to the GEOS NT SDK, you need two pieces of third-party software: a perl interpreter (a shareware copy is provided on the CD-ROM) and the Borland C/C++ compiler, version 4.0 or later. (Version 4.5 is recommended over 5.0; see "Compiling GEOS Applications," Section 4.)

2 Installation and Setup

This list summarizes the GEOS NT SDK installation steps. The text following the list explains each step in detail.

- Install the Borland C/C++ compiler (v4.5 recommended).

- Run **setup** from the GEOS NT SDK CD-ROM.
- Install perl (if not already present) by clicking on the "Perl Install" icon in the GEOS SDK Program Group. This icon points to **pcgeos\bin\Pw32i302.exe**.
- Run **sdkinstall** (the "Local SDK Install" icon) to complete the GEOS NT SDK installation and setup. (Run **pcgeos\bin\sdkinstall** if the icon is missing.)
- Modify the demo shortcuts in your GEOS SDK Program Group to use working directories from your private target tree. (Your personal copy of the target tree was made by **sdkinstall**.)
- Equalize foreground and background task priorities using the NT System control panel.
- Verify that environment and registry variables have been properly set.

If the GEOS SDK is installed on a shared network drive, you must perform the following setup on each NT development machine:

- Install Borland C++ and perl, or (if shared via network) add their **bin** directories to your PATH.
- Run **sdkinstall**.
- Create GEOS SDK Program Group shortcuts to **pcgeos\bin\swat.exe**, **pcgeos\bin\geossetup.exe**, and the target demos (**s.bat** and so forth). Make sure each shortcut uses the appropriate working directory and target in your target tree.
- Equalize foreground and background task priorities using the NT System control panel.
- Verify that environment and registry variables have been properly set.

2.1 Installation Details

Install the Borland C/C++ compiler (v4.5 recommended).

Installing the Borland C++ compiler prior to the GEOS NT SDK allows the SDK installer to automatically set the environment variable pointing to the directory where it finds the compiler. Be sure to install the 32-bit Windows compiler; it is part of the "typical" installation.

Run **setup** from the GEOS NT SDK CD-ROM.

Install the GEOS NT SDK by launching **setup.exe**, found in the CD-ROM's topmost directory. You will typically install the SDK under a directory whose last path element is PCGEOS, such as **C:\PCGEOS** for a local development tree or **S:\PCGEOS** (or other network drive) for a development tree served over a network to multiple workstations.

Install perl.

If you do not have a perl interpreter, install the shareware copy provided by running **pcgeos\bin\Pw32i302.exe** or clicking the "Perl Install" GEOS SDK Program Group icon. Leave checked the box to run **install.bat** after unzipping; reply "y" (the default) to the install options.

Run **sdkinstall**.

A perl script has been provided to complete the GEOS NT SDK installation steps required on each separate NT machine. Run **sdkinstall** (via the "Local SDK Install" icon or **pcgeos\bin\sdkinstall** from the command line) to complete the setup. It will:

- create a private development tree.
- create a private demo (target) tree.
- set your environment variables and update your NT configuration.
- set defaults in the variable registry.

The private directory trees you create with **sdkinstall** may be served over the network or kept on your local machine; in either case, the last path elements (under which the trees will be added) will typically be **pcgeos*privatedir*** and **pcgeos*privatedir*\target**. (See "Directory Structure" on page 13.)

Modify the demo shortcuts in your GEOS SDK Program Group to use working directories from your private target tree.

GEOS SDK Program Group icons are created automatically by the **setup.exe** installer. After you create a private target directory tree with **sdkinstall**, modify the working directory of

each demo icon so it points to this tree. (See "Target Directories" on page 15.)

When adding NT machines to a shared network SDK, you must add shortcuts to the GEOS SDK Program Group by hand. Refer to the icons on the original installation machine for guidance, or consult "Launching NT GEOS and Swat" on page 9.

Equalize foreground and background task priorities.
Disable priority boosting for foreground applications, as described in "NT Configuration" below.

Verify that environment and registry variables have been properly set.
Environment variables set or modified by **sdkinstall** are listed in "Environment Configuration" on page 7. You can modify variables in the registry with **GEOS Setup**, described in "Registry Configuration" on page 7.

2.2 NT Configuration

Task Priorities
Swat will not run properly unless all tasks (foreground and background) are set to run at the same priority. This setting is found in the **System** module of the NT Control Panel. Click on **Performance** (NT 4.0) or **Tasking** (NT 3.51) and set equal priorities (no foreground boost) for all tasks.

sdkinstall will complete the following items for you automatically; they are included here for reference only.

- Open File Count
  `files = 80` (or more) specified in **%SystemRoot%\system32\config.nt**.
- File System Driver
  `fs = os2{ec}.geo` specified under the `[system]` category in your demo's **.ini** file.

  `device = %SystemRoot%\system32\setver.exe` specified in **%SystemRoot%\system32\config.nt**.

  `setver loader.exe 9.0` and

```
setver loaderec.exe 9.0
```
executed in a command shell.

You can verify this setup by typing **setver** at a command line; entries for loader.exe and loaderec.exe should be listed. These settings persist across logins.

2.3 Environment Configuration

The compiling and linking tools use the following environment variables to find executables and include files. These variables are set from the **system** module of the control panel, not an autoexec file.

sdkinstall sets these environment variables for you; they are included here for reference only.

`ROOT_DIR` The root of the SDK installation, typically `C:\PCGEOS` (local development) or `S:\PCGEOS` (network development, substituting your particular network drive for `S:`).

`GOC_COMPILER_DIR`
The root of the compiler directory, such as C:\BC45.

`PATH` Your `PATH` must include entries for the **bin** directories of the Borland C++ compiler, PCGEOS, and your copy of perl. (More information on Borland C++ and perl is found in “Compiling GEOS Applications” on page 11.)

`CCOM` Alternate compilation command for **pmake**. This must be set to `@dosfront bcc` for versions of the Borland compiler prior to 5.0.

2.4 Registry Configuration

Swat uses the NT variable registry to find information about directory locations. **setup.exe** and **sdkinstall.cmd** set the registry to appropriate defaults; use **GEOS Setup** (in the GEOS SDK Program Group, or **geossetup** from the command line) to verify or modify the settings.

Swat (here and on its command line) prefers forward slashes in directory paths; backslashes are sometimes inadvertently interpreted as special characters (\tools as <TAB>ools, for instance).

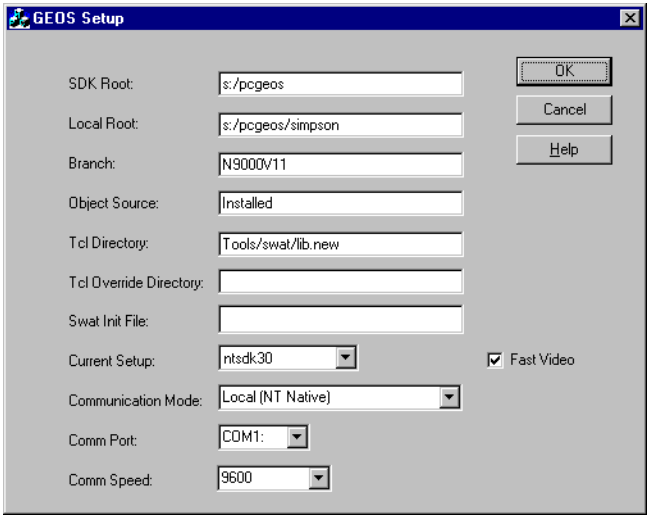


Figure 1 *GEOS Setup Window*
Use this editor to modify Swat's registry settings. Sample entries are for illustration only; see the text for proper settings.

SDK Root `C:/PCGEOS` (local) or `S:/PCGEOS` (network).
Can be any path, but the last element is typically PCGEOS. This is the directory where you installed the SDK.

Local Root `S:/PCGEOS/username`, for example.
Where Swat stores **.symdir** files; also the head of its search for an engineer's own **.sym** files. For **pmake** to run properly the penultimate element of *Local Root* must have the same name as the last element of *SDK Root*. (It need not be the actual *SDK Root* directory, however.) The last element may be named anything, but user names are typically used.

Branch `N9000V11`
or other platform as appropriate. Leave blank if your development tree does not have platform branches.

Object Source
`Installed`
Directory under *SDKRoot/Branch* heading the search for shared or SDK-supplied **.sym** files.

Tcl Directory
`Tools/swat/lib.new`
Location of Swat's **.tcl** files under *SDK Root*.

Tcl Override Directory
Location of additional **.tcl** files; typically blank.

Swat Init file
Any init file you would like Swat to execute upon startup. Typically blank or `C:/PCGEOS/SWAT.RC`.

Fast Video `Checked`
NT GEOS video mode selector. *Fast Video* artificially reduces the number of screen updates used by NT GEOS; therefore, you will see less flicker and faster execution response than on the actual device. Disable this option if you are testing animations or assessing relative flicker rates.

The **GEOS Setup** window also includes a selector for the communication mode. Choose `Local (NT native)` to instruct Swat to attach to your NT GEOS window. (Choose `Remote (Serial)` to instruct Swat to attach to an external device, with appropriate *Speed* and *Port* settings).

You may store and recall a number of different setups by using the *Current Setup* selector. The *Setup* names have no intrinsic meaning.

3 Launching NT GEOS and Swat

You will find example shortcuts for launching the NT GEOS demo in the GEOS SDK Program Group. You can define additional shortcuts to launch NT GEOS by right-clicking on the root window (NT 4.0) or selecting the menu item File -> New on the Program Manager (NT 3.51). Specify a command such as

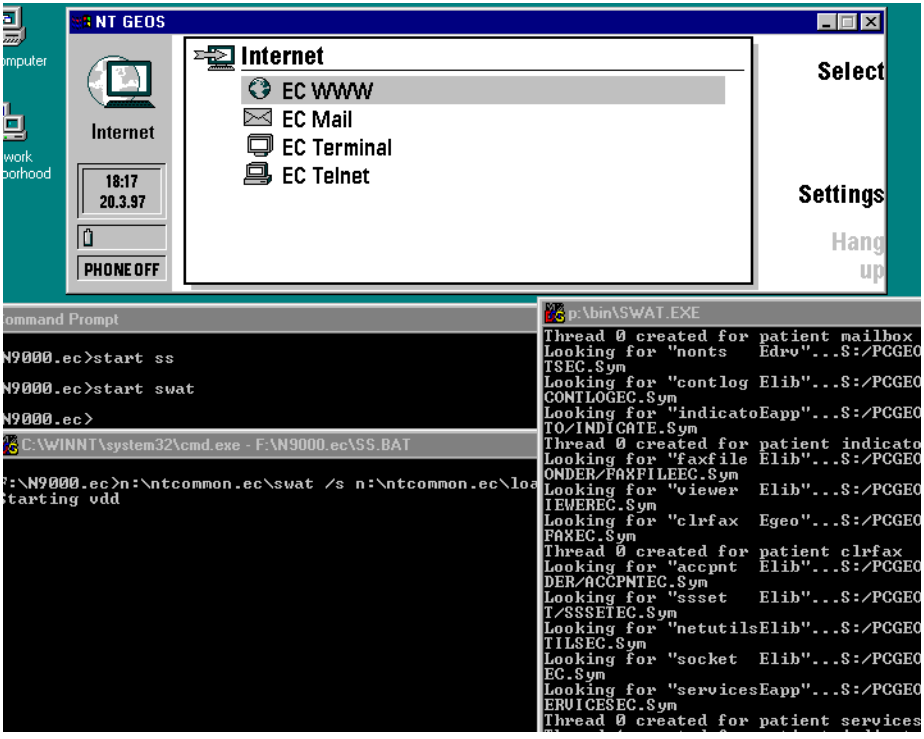


Figure 2 *Device Emulation (Demo) Example*

`c:\pcgeos\target\n9000.nc\localpc\swat.exe /s loader.exe` (modified as desired); you must also specify a working directory (under **target**\) corresponding to the simulation you wish this icon to run. For command examples, refer to the batch files described below.

To launch a target device window from the command line, change-directory to your target area (for example, **\pcgeos\target\n9000.ec\localpc**) and type **start s** at the prompt. (**s.bat** is supplied in the target directory templates, and **start** tells NT to run the command in a new window.) **s.bat** is

a batch file which executes the Swat stub; **ss.bat** forces the demo to wait for a host Swat debugger to attach. Execute **start s** (or **ss**) in your non-EC directory to run the non-EC version of the Swat stub. To run the non-Swat demo, use **start geos**, which executes **loader{ec}.exe** without the Swat stub.

In summary, each of these demos is available in both the EC and non-EC target directories:

```
ss.bat      Demo with Swat stub, wait for host Swat
s.bat       Demo with Swat stub, do not wait
geos.bat    Demo without Swat
```

Launch the host Swat debugger from an icon in the GEOS SDK Program Group or from the command line with **swat** (or **start swat**). (Be careful not to run a Swat stub if your current directory is a target directory; you want to run **swat.exe** from **pcgeos\bin**.) Debugging with Swat in the NT environment is the same as that in the old DOS one; use **^C** and **c** to interrupt and continue execution, **att** and **det** to attach and detach, and so forth.

The Swat window size may be increased by modifying the window-size property of your command windows in general or your Swat shortcut in particular. Any change made to a window *after* Swat is launched will not be recognized.

You can instruct Swat to attach to an external Communicator device or PC emulator by running **GEOS Setup** prior to launching Swat and selecting the serial *Communication Mode* and an appropriate *Comm port*.

You can stop NT GEOS by entering **detach** or **quit** when Swat is attached. Otherwise, you can stop the demo by clicking the kill (**X**) button on the right of the titlebar, double-clicking the top-left titlebar menu, or using NT's task manager.

4 Compiling GEOS Applications

To compile GEOS C code, you need the following third-party software installed on your system:

- Borland C/C++ Compiler

- Perl Interpreter

Your PATH variable must include the Borland compiler's **bin** directory and the directory containing the perl interpreter. If you installed Borland C++ after the NT SDK or if you set up a network environment, you may need to set the environment variable GOC_COMPILER_DIR to reflect the compiler's location. Refer to "Environment Configuration" on page 7.

Borland C++ is a separate product on its own CD-ROM. We strongly recommend that you install version 4.5 of the compiler, included with most distributions of version 5.0. A change in the compiler's handling of #pragma options creates errors (not reported at compile time) in geodes compiled with version 5.0. You must install the 32-bit Windows compiler, a part of the "typical" installation.

A shareware version of the perl installer has been included with the NT SDK; refer to "Installation and Setup" on page 3 for installation instructions.

All of the tools needed to compile geodes (goc, glue, etc.) have been included in the **\pcgeos\bin** directory. Use **mkmf** and **pmake** to compile, as described in the chapter "Using Tools" of the Tools book. (**pmake depend** is not needed in the NT environment.)

After compiling, copy resulting geodes directly to your target tree from the command line or use Swat's **send** command after attaching. **world\extrapps** is the appropriate demo directory for extras on the Nokia 9000 Communicator.

4.1 Changes From the DOS SDK

If you are familiar with GEOS development in the DOS environment, note the following changes:

- You need not run **pmake depend** after **mkmf**; use it only to re-make dependencies when you change your include-file statements.
- You may use DOS **copy** to put compiled geodes in your target tree, although Swat's **send** command works as before.
- Subdirectories under the main include-file directories must be explicitly stated, as in `#include <Ansi/string.h>`.

- The “pause” and “break” keys do not stop the demo; kill the GEOS window using the menubar or use Swat’s **detach** or **quit** command.
- The development directories have been modified; see “Directory Structure” below.

5 Directory Structure

The directory structure in the NT SDK differs significantly from that of the DOS SDK. The change reflects standardization across platforms and facilitates multi-engineer development in a network environment.

5.1 Branches

Every device that runs GEOS contains a specific version of the kernel, applications, and various ancillary libraries and drivers. The directory path that contains the specific version of the device software is called a Branch. (The “Trunk” is a collection of all the files from all the different branches. As a developer, you will never need to utilize any files on the Trunk.)

Branch directories are located directly under the PCGEOS directory (*i.e.*, the main SDK directory) and were developed as a way to “lock in” a device-specific image of the development system. For example, the N9000V11 branch contains all the header and symbol files for Version 1.1 of the Nokia 9000 SDK. When you install an SDK for another GEOS device, you will find a new branch directory specific to that device under the PCGEOS directory.

Figure 3 shows the directory layout of the Nokia 9000 SDK (Version 1.1). This directory tree contains two Branch directories: `N9000V11` and `{Working_Branch}`. (`{Working_Branch}` is your personal development area; see section 5.4 on page 16 for a complete explanation of Working Branches)

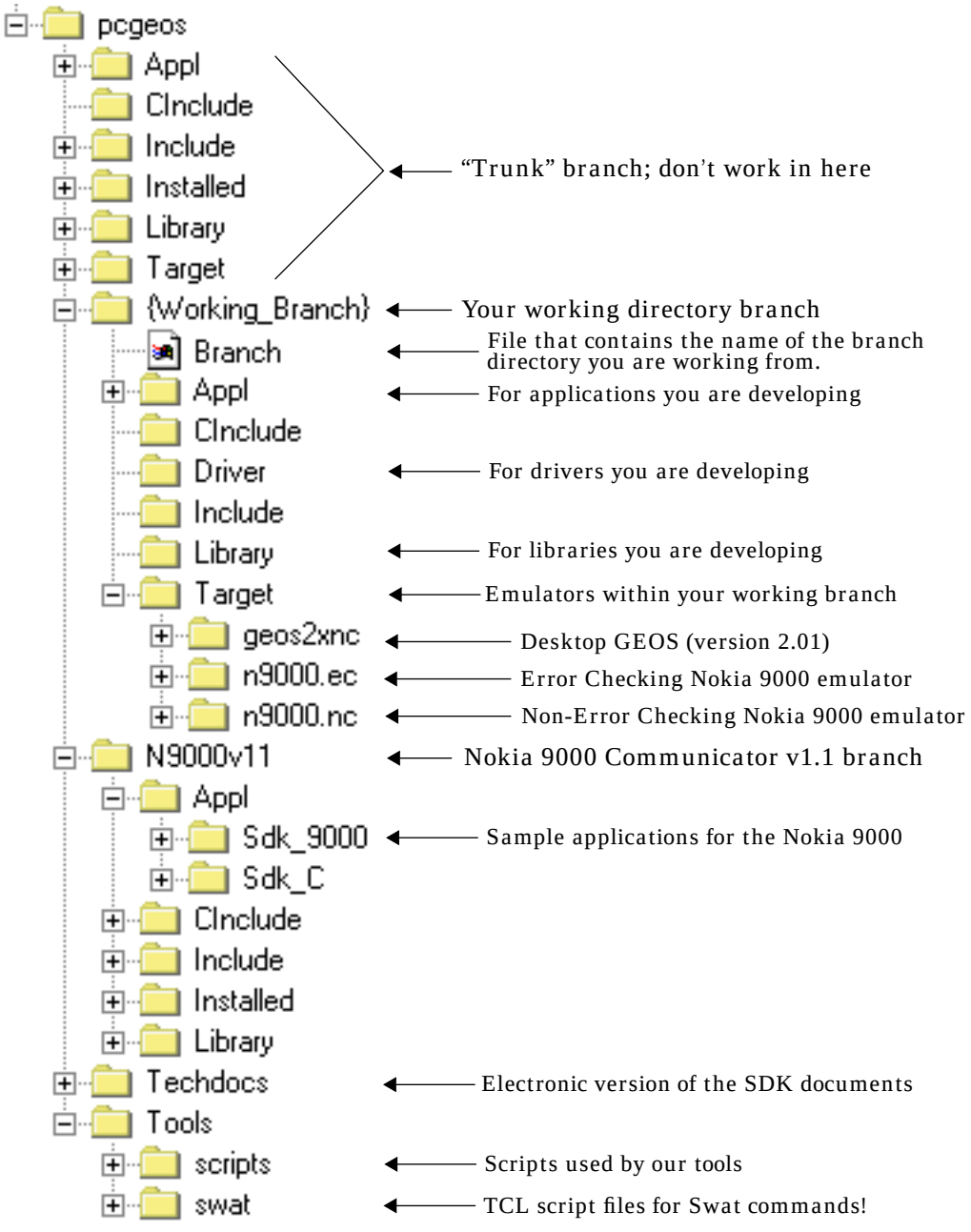


Figure 3 *Development Directory Tree*

5.2 Main Directories

Each Branch contains the following directories:

`CInclude` Contains all the *.H and *.GOH header files you need to compile GEOS applications written in GOC.

`Include` Contains all the *.DEF and *.UIH header files you need to compile GEOS applications written in ESP (assembly).

`Installed` Contains all the *.LDF (*i.e.*, link definition files) files (used by Glue, our linker) and symbol files of all the GEOS applications, libraries, and drivers (used by Swat, our debugger).

`Appl` Contains a subdirectory holding your application's source code; for instance, your CardGame application might reside in: **appl\cardgame\cardgame.goc**

Also contains the following subdirectories:

SDK_C which holds general purpose sample applications; for example, the huge array sample application lives in: **appl\sdk_c\harray\harray.goc**

SDK_* which holds device-specific sample applications (where * refers to the specific device); for example, the dialog box sample application for the Nokia 9000 Communicator lives in: **appl\sdk_9000\dialogs\dialogs.goc**

`Library` Contains a subdirectory holding your library's source code; for example, the new utility library you're writing would live in: **library\utility\utility.goc**

`Driver` Contains a subdirectory holding your driver's source code; for example, the new Zippy driver you're writing would live in: **driver\zippy\zippy.goc**

5.3 Target Directories

In addition to those directories listed above, your Working Branch contains a Target directory. This directory contains the executable and data files comprising an emulation (or "demo") of the actual GEOS device. The Target subtree mirrors the file system layout of the actual device.

Be sure to update the working directories of your GEOS SDK demo shortcuts so that they point to the Target directory in your Working Branch; for example, the shortcut to the error-checking demo for the Nokia 9000 Communicator should be **\pcgeos\username\target\n9000.ec**.

Network installations can make use of shared geode libraries if desired. When geodes are invoked during emulation, the current working (target) directory tree will be searched first, followed by trees specified by the `top` keyword under `[paths]` in your **.ini** file. For example, all of the geodes under the **system** subdirectory could be stored in a central, shared location in order to save disk space or allow easy updates. You can supersede shared geodes in your demo by putting custom versions (of the same name) in your private **system** subdirectory.

Each target directory has a **geos{ec}.ini** file. It can refer to files of standard **.ini** settings under its `[paths]` category; it also contains superseding information necessary for running NT GEOS. When editing **.ini** files, take care to end each line with a `CR-LF` pair or use an editor (such as DOS **edit**) that does so transparently. More information on **.ini** files in general is found in the Tools book.

5.4 Developing In Your Working Branch

The Working Branch is your private development area. By doing all your work in your Working Branch (also called your *Local Root*), the Branch files can remain untouched.

You specify the directory for your Working Branch during the Local SDK Install program. The default is your login name. For example, if your login is jdoe, then your Working Branch directory is **c:\pcgeos\jdoe**. (You should also make sure that the demo shortcuts point to the Target directory in your Working Branch.)

Within your Working directory is a file called BRANCH. This file specifies which Branch directory the tools will use when compiling and linking your source code. The tools will first look for files in your Working Branch. Whatever files it does not find in your Working Branch, it will look for in the Branch specified by the BRANCH file. Thus, you need only keep the file or files that you are working on in your Working Branch and system files (such

as header files) can remain on the device Branch. So, for example, a typical development sequence might look like this:

- You want to add another feature to the Zippy driver so you copy (or "check out" if you have a version control system) the zippy.goc file from the Branch to your Working Branch.
- You edit the file and make your changes.
- You compile the modified file.
- You find and fix all the bugs.
- You "install" the modified file using version control or by simply copying it back to the Branch.

To specify which Branch the compiler and linker should use, place the name of the Branch directory in the BRANCH file. (Currently, setting the Branch field in GEOS Setup only affects which branch Swat uses; you must edit the actual BRANCH file yourself.) Initially, your BRANCH file contains N9000V11 since that is the only Branch currently available.

5.5 Multi-Developer Projects

The system of Branch files is most helpful when there are multiple developers working on the same project. Each developer would maintain her personal working files in her own Working Branch while shared files would remain on the device Branch. As development progresses, each developer can "install" files to the Branch directory as she reaches stable milestones thereby allowing her co-workers to use the "installed" versions of her files. (Note: "installing" can be as simple as copying the files by hand or can be handled by a version control system.)

6 Troubleshooting

If you are having trouble starting the simulation or attaching Swat, consult the table below. The most common known difficulties are listed.

GEOS window dies immediately.

Run the EC version and attach Swat. Does it die with the error LS_CANNOT_LOCATE_KERNEL?
*The working directory of the NT GEOS window must be a target directory with an **.ini** file and device subdirectories.*

Does Swat report FSHOF_SET_FILE_NAME_PASSED_TO_SKELETON_DRIVER, ATTEMPTED_TO_PERFORM_UNSUPPORTED_FUNCTION_BEFORE_APPROPRIATE_IFS_DRIVER_WAS_LOADED, or TOO_MANY_PRIMARY_FSDs?
*There is a problem with the file system driver. Check **.ini** file settings and **setver** listing, referring to "NT Configuration" on page 6.*

GEOS is not in a window.

It should have a title bar and look like a window.
*Verify that you have a text loader (**loader{ec}.exe**).*
*Verify that **geosvdd.dll** is in your path.*
Verify that you are using an NT video driver under `[screen 0]` *in your **.ini** file (probably **simp4bi{t/e}**).*

GEOS cannot launch, reporting: `Can't open named pipe`

Are you already running NT GEOS?
Only one copy can run at a time.

Swat cannot connect, reporting: `Can't contact GEOS using the Named Pipe: \\.\pipe\swatpipe`

Are you running the demo with a Swat stub (**s.bat** or **ss.bat**)? Did you immediately try to attach?
If you are running the correct demo, try to attach again. You may have attempted to attach before the Swat stub was ready.

The keyboard doesn't work.

Is the NT GEOS window active? It should have a blue title bar.
Click in the window to make it active.

GEOS isn't running a window.
*Check your **.ini** file (and [paths] referents) for correct keyboard setting (or* `pini keyboard` *in swat); you must use the **gdikbd** keyboard driver.*

The keyboard seems to fail suddenly.
Ctrl or Alt keys may be "stuck" in GEOS. Hit Ctrl, Alt, Caps Lock, and Shift a few times to clear them.

Extended-character-set dialog will not come up.
Verify that `altGr = true` *under category* `[GDI keyboard]` *is set in your* ***.ini*** *file (or referent).*

Swat's **srcwin** command fails, reporting: `Window too small`
The default NT command-window size is too small for the default srcwin. There are two fixes:
(1) Reduce the ***srcwin*** *buffer size to match your window size with the Swat command* `srcwin 20.`
(2) Restart Swat from a larger window. (Size changes during a session are not recognized by Swat.) Your default window size can be increased by changing the window-size property of your command windows, or just your Swat shortcut.

Swat does not set mouse-click breakpoints.
Make sure quick-edit mode is off in your Swat window.

GEOS hangs until Swat window selected.
Have foreground/background task priorities been set equal?
Check under the ***System*** *module of the Control Panel.*

GEOS hangs or **.ini** file corrupt.
Run the EC version and attach Swat. Is the **.ini** file corrupt?
Open and re-save it using NT or DOS ***edit*** *to restore* `CR-LF`*'s.*

GEOS does not emulate sounds properly.
This is a known sound bug—disable sound with `sound = off` *under* `[UI]` *in your* ***.ini*** *file.*

pmake fails, reporting: `Couldn't find geos.mk.`
Set environment variable ROOT_DIR *(to* C:\PCGEOS*, e.g.).*

pmake fails, reporting: `Couldn't find stddef.h`
Set environment variable GOC_COMPILER_DIR *(to* C:\BC45*, e.g.).*


GEOS NT
Development